

Manoj Kumar Thapa

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UK Graduate Route visa valid to 2028 – no sponsorship required

EDUCATION

MSc Artificial Intelligence, Aston University, UK – Distinction (70.37%) Jan 2025 – Jul 2026

BE Computer Science, Dr. Ambedkar Institute of Technology, India – CGPA 9.45/10 Aug 2017 – Sep 2021

PUBLICATION

Vinutha H., **M. K. Thapa**, P. Pandey, N. Pun, N. Paudel. *Indian Currency Recognition for Visually Impaired using Deep Learning Technique*. IJIRCCE, 9(7), July 2021. DOI: 10.15680/IJIRCCE.2021.0907159 · [PDF](#)

PROJECTS

FruitGuard AI – MSc Dissertation | *PyTorch, YOLOv8, ONNX Runtime* | [Demo](#) Aug 2025 – Apr 2026

- Fine-tuned a YOLOv8 model with a two-phase transfer-learning pipeline to **96.3%** classification accuracy (52 of 54) from only **207** labelled images, a **34x** reduction against the **7,108** images used for pre-training.
- Beat the COCO-pretrained baseline by **+6.03 pp** mAP@0.5:0.95 (0.5854 to 0.6457) and **+3.12 pp** precision.
- Optimized inference with ONNX Runtime to real-time **73 FPS**, deployed as a live Hugging Face demo.

Planck AI – Agentic Research Assistant | *Python, FastAPI, React, Docker* | [GitHub](#) · [Demo](#) Sep 2025 – Nov 2025

- Built an LLM agent in Python without an agent framework, orchestrating 4 tools (web search, code execution in 7 languages, PDF/URL reading, image analysis) across up to 8 planning steps.
- Designed mid-run failover across Groq, Gemini and NVIDIA behind one OpenAI-compatible interface, sustaining **30/30** task completion under fault injection with the primary provider disabled on every call.
- Built an evaluation harness with a 30-question golden set and a rubric committed to Git before the first run, measuring **80%** correct tool selection, **2.7 s** median latency and no incorrect answer among the **20** tasks the rubric could grade.

CogniGraph – Graph RAG Explorer | *Python, ChromaDB, NetworkX, OCR* | [GitHub](#) · [Demo](#) Jun 2025 – Jul 2025

- Built a knowledge-graph RAG app that turns documents, including scanned PDFs via OCR, into an interactive 3D graph and answers only from passages retrieved from a ChromaDB vector database.
- Rebuilt the evaluation set after finding self-written test questions overstated accuracy by **35 points** (90% to 55% hit@1), then benchmarked the corrected set at **95% hit@3** against **80%** for a keyword baseline on the same questions.
- Proved the embedding model silently discards text past **~1,400 characters**, showed the sweep favouring 2,000-character chunks was confounded by corpus size, and set chunk size to **1,000**, the largest that fits the window.

NeuroArc – AI Job Application Assistant | *FastAPI, React, Groq API, OCR* | [GitHub](#) · [Demo](#) Nov 2025 – Dec 2025

- Built an end-to-end assistant that parses CVs (PDF/DOCX, OCR fallback), pulls live UK jobs from the Reed API, scores CV-job fit and exports a tailored CV as PDF.
- Stopped the model inventing its own evidence: it returns requirement-evidence pairs as structured JSON, and code discards any requirement whose quoted evidence is not in the CV before computing the weighted score.

EXPERIENCE

Mentor, MSc Artificial Intelligence, Aston University | *Birmingham, UK* Aug 2025 – Jan 2026

- Mentored **5** MSc AI students in Python and machine learning, one-to-one, every week.
- Focused sessions on the three blockers that recurred most: setting up Python environments, debugging training loops that would not converge, and choosing a simple baseline before a complex model.

Software Engineer, Accenture | *Bengaluru, India* Oct 2021 – May 2022

- Cut p95 REST API latency from roughly **850 ms to 510 ms** across **18 endpoints**, a **40%** reduction, by eliminating **6 N+1** query sites and rewriting the SQL behind them in Java Spring Boot.
- Owned the business logic and the PostgreSQL/MySQL data-access layer for that service, shipping production REST APIs in a **7-engineer** delivery team on 2-week sprints.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript
LLMs & Agents: Tool/function calling, agent loops, LangGraph, RAG, prompt engineering, structured JSON output, multi-provider LLM fallback (Groq, Gemini, NVIDIA; OpenAI-compatible APIs)
Retrieval & Data: ChromaDB (vector database), embeddings, semantic search, OCR, PostgreSQL, MySQL
ML & Computer Vision: PyTorch, YOLOv8, transfer learning, fine-tuning, ONNX Runtime, quantization
Evaluation: Golden-set design, fault injection, baseline controls, latency benchmarking
Backend & Frontend: FastAPI, Pydantic, Spring Boot, REST API design, React, Tailwind CSS
Tools & Deployment: Docker, Docker Compose, Git, Hugging Face Spaces, Render

CERTIFICATIONS

AI Agents Intensive (Google, Kaggle) | Machine Learning Specialization (Stanford) |
Deep Learning Specialization (DeepLearning.AI)